General & Diagnostic Microbiology

Q1: Draw and label a typical bacterial cell. What are the cell envelopes? Mention the essential and non-essential structure of bacteria. Mention the function of capsule and cell wall. Enumerate 5 capsulated bacterias.

Q2: What are flagella and pili? Classify flagella and pili with examples. Mention the difference between them. State the function and importance of flagella. Mention 5 flagellated bacteria.

Q3: Classify bacteria on the basis of morphology with example. Mention the name of anaerobic bacteria.

Q 4: What is bacteria generation time? Give example.

Q5: Define staining. Mention two important bacterial staining methods. Discuss the procedure of gram and AFB Staining.

Q6: Define sterilization, disinfection and antisepsis. Classify sterilization. Write down the use and difference between hot air oven and autoclave. Write down the merits and demerits of a hot air oven.

Q 7: What is culture and culture media? Classify media on the basis of consistency and constituent with examples. Mention two enrich and two selective media. Mention the advantages and disadvantages of liquid media.

Q 8: Describe the preparation of blood agar media and mention its use. Name the organisms that grow in the following media: i) Chocolate agar, ii) Mac Conkey agar, iii) Blood tellurite media.

Q9: What is an antimicrobial agent? Mention the difference between antimicrobial agents with their site of action. Discuss in short, the mechanism of action of penicillin. Mention the difference between gram positive and gram negative bacterial cell wall. What is MIC and MBC?

Q10: Define toxin and toxoid. Write down the difference between exotoxin and endotoxin. Mention 5 exotoxin and endotoxin producing bacteria.

Q11: Mention the different types of microscopes, give an example. Enumerate 2 bacteria that can be seen by dark ground microscopy (DGM). (Note: This question was present but not answered in the notes.)

Q12: What is Selective toxicity and drug resistance? Name 4 beta-lactum drugs and 4 drugs that act by inhibiting protein synthesis. How bacteria develop resistance to penicillin? Describe the method of development of drug resistance.

Question 13: What is growth and growth curve of bacteria? Draw and label a bacterial growth curve. Mention in which phase exotoxin, endotoxin and spores are produced.

General & Diagnostic Microbiology

Question 1

Draw and label a typical bacterial cell. What are the cell envelopes? Mention the essential and non-essential structures of bacteria. Mention the function of the capsule and cell wall. Enumerate 5 capsulated bacteria.

Diagram: Structure of Bacteria
(Diagram showing: Flagella, Ribosome, Nucleoid DNA, Cytoplasm, Attachment Pili, Sex Pilus, Capsule, Cell Wall, Cell Membrane)

Cell Envelope
The cell envelope is a combination of the cell membrane, cell wall, and outer membrane, or capsule.

Essential and Non-essential Structures

- **Essential Structures:**
    1. Cell wall
    2. Cell membrane
    3. Ribosome
    4. Nucleoid
    5. Mesosome
    6. Periplasm
- **Non-essential Structures:**
    1. Flagella
    2. Pili
    3. Capsule
    4. Glycocalyx
    5. Spore
    6. Plasmid

Function of Cell Wall

1. Provides structural support.
2. Maintains the characteristic shape of the cell.
3. It is the site of action of penicillins and cephalosporins.
4. It gives osmotic protection and prevents lysis of the cell.
5. It mediates antigenicity.
6. It helps in cell division.

Function of Capsule

1. The capsule may play a role in the adherence of bacteria to human tissue, which is an important initial step in causing infection.
2. **Antiphagocytic:** Prevents phagocytes from engulfing the bacteria.
3. Specific identification of an organism can be made by using anti-serum against the capsular polysaccharide (Quellung reaction).
4. Capsular polysaccharides are used as the antigens in certain vaccines. For example, the purified capsular polysaccharide vaccine of *S. pneumoniae*.

5 Capsulated Bacteria

1. *Streptococcus pneumoniae*
2. *Klebsiella pneumoniae*
3. *Haemophilus influenzae*
4. *Neisseria meningitidis*
5. *Cryptococcus neoformans (**fungus (yeast)**, not a bacterium)*
6. *Salmonella*
7. *E. coli*

Question 2

What are flagella and pili? Classify flagella and pili with examples. Mention the difference between them. State the function and importance of flagella. Mention 5 flagellated bacteria.

Flagella
It is a long, thick, curved whip-like appendage, embedded in the cell membrane. It is made of flagellin protein.

Pili
Short, straight, thin, hair-like filament that extends from the cell surface. Composed of pilin protein. Present mainly in Gram-negative bacteria.

Classification of Flagella

1. **Monotrichous:** Single flagellum at one pole. (e.g., *Vibrio cholera*)
2. **Amphitrichous:** Flagella at both poles. (e.g., *Campylobacter jejuni*)
3. **Lophotrichous:** Tuft of flagella at one pole of the bacteria. (e.g., *Spirillum*)
4. **Peritrichous:** Flagella all over the bacteria. (e.g., *Salmonella*)

Classification of Pili

1. **Ordinary pili / Short attachment pili**
2. **Sex pili / Long conjugation pili**

Difference between Flagella and Pili

Traits	Flagella	Pili
Appearance	Long, thick, curved whip-like appendages.	Short, straight, thin hair-like filaments.
Composition	Composed of flagellin protein.	Composed of pilin protein.
Types	Four types.	Two types.
Function	It helps in the motility of bacteria.	Attachment and genetic transfer (conjugation).

Function of Flagella / Importance

1. Organ of locomotion (Motility).
2. Some species of bacteria are identified in the clinical laboratory by the use of specific antibodies against flagellar proteins. It is antigenic, termed as the 'H' Antigen.

5 Flagellated Bacteria

1. *Vibrio cholera*
2. *Campylobacter jejuni*
3. *Helicobacter pylori*
4. *Treponema pallidum*
5. *Salmonella enterica*

Question 3

Classify bacteria on the basis of morphology with examples. Mention the name of anaerobic bacteria.

Morphological Classification of Bacteria

1. **Cocci (Round shaped):** *Streptococci*, *Staphylococci*.
2. **Bacilli (Rod shaped):** *E. coli*, *Salmonella*.
3. **Coccobacilli (Ovoid):** *Bordetella pertussis*, *Haemophilus influenzae*.
4. **Filamentous (Branching):** *Actinomycetes*, *Nocardia*.
5. **Curved (Comma shaped):** *Vibrio cholerae*, *Helicobacter pylori*.
6. **Spiral:** *Treponema*, *Borrelia*, *Leptospira*.

Anaerobic Bacteria

- **Gram-negative:**
    1. *Bacteroides*
    2. *Fusobacterium*
    3. *Porphyromonas*
    4. *Prevotella*

- **Gram-positive:**
    1. *Actinomyces*
    2. *Clostridia*
    3. *Peptostreptococcus*
    4. *Cutibacterium*

Question 4

What is bacteria generation time? Give an example.

Bacteria Generation Time
It is the time required for bacteria to become double in number under optimum concentration of growth. It is also known as doubling time.

Examples:

1. *E. coli* → 20 mins
2. *Tubercular bacilli* → 18-20 hours
3. *Mycobacterium leprae* bacilli → 20 days

Question 5

Define staining. Mention two important bacterial staining methods. Discuss the procedure of Gram and AFB Staining.

Staining
Staining is the technique of visualization of a micro-organism by enhancing the contrast of the image by applying natural or artificial organic compounds.

Two important bacterial staining methods:

1. Gram stain
2. Ziehl-Neelsen stain / AFB stain

Gram Staining

- **Principle:** Gram staining depends on the cell wall structure and cytoplasmic membrane. It separates most bacteria into two groups, Gram-positive bacteria and Gram-negative bacteria. Gram-positive bacteria stain blue/purple/violet and Gram-negative bacteria stain red/pink.
- **Procedure:**
    1. Make a smear with the supplied specimen on a clean, grease-free slide and dry in air.
    2. Fix the smear by slowly passing the smear slide over a flame 3-4 times.
    3. **Primary Staining:** Place the smear on a staining rack and cover it with crystal violet and wait for 1-2 min then wash with water.
    4. **Mordanting:** Cover the slide with Lugol's iodine and wait for 1-2 min, then wash.
    5. **Decolorization:** Decolorize by acetone for 2-3 seconds followed by a wash with water. Repeat until the slide is decolorized.
    6. **Counter Stain:** Counterstain by carbol fuchsin and wait for 1-2 minutes.
    7. Wash and dry in the air.
    8. Examine under an oil immersion lens.

- **Interpretation:**
    1. Organisms show blue/purple/violet color — Gram positive.
    2. Organisms show red/pink color — Gram negative.

AFB Stain (Acid-fast bacilli)

- **Principle:** The cells of certain organisms containing long-chain fatty acid (mycolic acid) give them a coat impervious to crystal violet or other basic dyes. Heat or detergent must be used to allow penetration of the primary dye.
- **Procedure:**
    - The dried smear is heat-fixed.
    - **Primary Stain:** The smear is covered with hot carbol fuchsin and allowed to remain on the slide for 5 minutes.
    - The stain is washed off with clear water.
    - **Decolorization:** The smear is treated with 20% H_2SO_4 for decolorization and wait for 1 min.
    - **Counter Stain:** Counterstain is done by applying methylene blue or malachite green and waiting for 1-2 minutes then wash with water.
    - Examine under an oil immersion lens.

- **Observation:**
    - Acid-fast bacilli appear red in a blue or green background of pus cells or epithelial cells.

Question 6

Define sterilization, disinfection and antisepsis. Classify sterilization. Write down the use and difference between hot air oven and autoclave. Write down the merits and demerits of a hot air oven.

Sterilization
Freeing of an article from all microorganisms like viruses, bacteria, fungus and including spores. This is an absolute term and can be done by physical or chemical process.

Disinfection
A process that eliminates many or all pathogenic microorganisms but not spores from inanimate objects.

Antisepsis
It is a procedure which inhibits or destroys microorganisms in living tissue.

Classification of Sterilization

1. **Physical:**
    - **A. Heat:**
        - Dry heat - Red heat, Flaming, Incineration, Hot air oven
        - Moist heat - Autoclave (Above 100°C), Boiling / Tyndalization (At 100°C), Pasteurization (Below 100°C)
    - **B. Radiation:** a) Ionizing radiation, b) Non-ionizing radiation
    - **C. Filtration:** a) HEPA filters, b) Berkfeld
2. **Chemical:**
    - **A. Liquids:** a) Alcohol, b) Aldehydes, c) Heavy metals, d) Dyes
    - **B. Gaseous:** a) Formaldehyde, b) Ethylene oxide, c) Plasma

Use of Hot Air Oven

1. All glassware like - Glass syringe, Petridish, Test tube, Flasks, Pipettes etc.
2. Powders, Vaseline, Oils.

Use of Autoclave

1. Sterilization of culture media except blood, serum, egg containing media.
2. Swabs for use in microbiology.
3. Surgical dressing, linens, gloves etc.
4. Blunt surgical instruments.

Difference between Hot Air Oven and Autoclave

Hot Air Oven	Autoclave
1. Requires more time.	1. Requires less time.
2. Requires high temp.	2. Requires low temp.
3. Latent heat is not liberated.	3. Latent heat is liberated.
4. Has less penetrating power.	4. Has more penetrating power.
5. Water-containing substances cannot be sterilized. Cotton gets charred.	5. Suitable for sterilization of culture media (except blood, serum, egg-containing media).

Merits of Hot Air Oven

1. Glasswares like petridish, swab, sticks, and glass syringe can be sterilized.
2. Powders, fats, oils etc. can be sterilized.
3. Sharp cutting instruments can be sterilized.

Demerits of Hot Air Oven

1. Needs more temperature.
2. Time consuming.
3. Plastic material can't be sterilized in this process.

Question 7

What is culture and culture media? Classify media on the basis of consistency and constituent with examples. Mention two enriched and two selective media. Mention the advantages and disadvantages of liquid media.

Culture
It is the artificial process of cultivation of microbes by providing the proper food and environmental conditions.

Culture Media
Artificial food for bacterial growth and multiplication in the laboratory.

Classification of Media on the basis of Consistency

1. **Liquid media (Contains no agar):** i) Nutrient broth, ii) Alkaline peptone water, iii) Tryptic soya broth (TSB)
2. **Semi-solid media (Contains 0.2-0.5% agar):** i) Motility Indole Urea (MIU), ii) Soft agar media
3. **Solid media (Contains 1-2% agar):** i) Nutrient agar media, ii) Blood agar media, iii) Chocolate agar media, iv) MacConkey's agar media

Classification of Media on the basis of Constituents

1. **Simple/Basal media:** a) Nutrient broth, b) Peptone water
2. **Enriched media:** a) Blood agar, b) Chocolate agar
3. **Selective media:** a) MacConkey agar, b) Lowenstein Jensen media
4. **Indicator media:** a) Blood agar
5. **Enrichment media:** a) Tryptic soya broth
6. **Transport media:** a) Cary-Blair media
7. **Storage media:** a) Egg-saline media

Two Enriched Media:

1. Blood agar media
2. Chocolate agar media

Two Selective Media:

1. MacConkey's agar media
2. Lowenstein Jensen media

Advantages of Liquid Media

1. A small amount of bacteria in the specimen is sufficient.
2. Rapid growth of bacteria.
3. Biochemical tests can be performed.
4. Large inoculation can be tested.
5. Helps in transport of a specimen.

Disadvantages of Liquid Media

1. Identification of bacteria by colonial morphology is not possible.
2. Individual organisms cannot be isolated or separated from a specimen of mixed infection.
3. Antibiotics sensitivity tests cannot be performed.
4. No colony formation.

Question 8

Describe the preparation of blood agar media and mention its use. Name the organisms that grow in the following media: i) Chocolate agar, ii) MacConkey agar, iii) Blood tellurite media.

Preparation of Blood Agar Media

1. Sterilize the agar medium by autoclaving at 121°C for 15 min.
2. Transfer to a 50°C water bath.
3. When the agar has cooled to 50°C, add aseptically the sterile blood and mix gently but well. Avoid forming bubbles.
4. The blood must be allowed to warm to room temperature before being transferred to the molten agar.

Use of Blood Agar Media

1. To identify microorganisms.
2. Differentiation of bacterial species is done by the hemolysis property of the bacteria.
3. Three types of hemolysis occur here, α, β, and γ.

Name of the organisms that grow in the Chocolate Agar Media

1. *Streptococcus pneumoniae*
2. *Haemophilus influenza*
3. *Neisseria meningitidis*
4. *Neisseria gonorrhoeae*

Name of the organisms that grow in the MacConkey's Media

1. *Salmonella*; 2. *Klebsiella*; 3. *E. coli* etc.

Name of the organism that grows in the Blood Tellurite Media
Corynebacterium diphtheriae (black colony)

Question 9

What is an antimicrobial agent? Mention the different antimicrobial agents with their site of action. Discuss in short, the mechanism of action of penicillin. Mention the difference between gram positive and gram negative bacterial cell wall. What is MIC and MBC?

Antimicrobial Agent

Antimicrobial agents are drugs that either kill or inhibit the growth of microbes within a host and therefore are effective in the treatment of infectious agents.

Different antimicrobial agents with their site of action

1. **Inhibition of the cell wall synthesis:**
    - A. Cross-linking of peptidoglycan synthesis inhibitors:
        - i) Beta-lactam drugs - Penicillins, Cephalosporins, Carbapenems, Monobactams
        - ii) Vancomycin
    -
    - B. Other steps of peptidoglycan synthesis inhibitors:
        - i) Non-beta lactam drugs - Bacitracin, Cycloserine
    -
2. **Inhibition of protein synthesis:**
    - A. 50S ribosomal subunit: i) Chloramphenicol, ii) Clindamycin, iii) Erythromycin, iv) Linezoid, v) Azithromycin
    - B. 30S ribosomal subunit: i) Aminoglycosides, ii) Tetracyclines
3. **Inhibition of nucleic acid synthesis:**
    - A. Nucleotide synthesis: i) Sulfonamides, ii) Trimethoprim
    - B. DNA synthesis: i) Quinolones
    - C. mRNA synthesis: i) Rifampin
4. **Alteration of cell membrane function:**
    - A. Antibacterial activity: i) Polymyxin, ii) Daptomycin
    - B. Antifungal activity: i) Amphotericin B, ii) Nystatin,

      iii) Ketoconazole **(inhibition of ergosterol synthesis)**

5. **Other mechanism of action:**
    - A. Isoniazid, B. Metronidazole, C. Ethambutol, D. Pyrazinamide

Mechanism of action of penicillin

1. By binding with penicillin-binding protein (PBP), which acts as transpeptidase enzymes.
2. By inhibiting transpeptidase - The enzymes that catalyze the final cross-linking step in the synthesis of peptidoglycan.
3. By activation of autolytic enzymes (Murein hydrolase) which degrade the cell wall (Peptidoglycan).

Difference between gram positive and gram negative bacterial cell wall

Traits	Gram-positive Cell Wall	Gram-negative Cell Wall
1. Peptidoglycan	Thicker, 40 sheets	Thinner, 1-2 sheets
2. Teichoic acid	Present	Absent
4. LPS (Lipo-polysaccharide)	Absent	Present
5. Periplasmic space	Absent	Present
6. Porin Protein	Absent	Present
7. Lipoprotein	Absent	Present

MIC (Minimal inhibitory concentration)
is the lowest concentration of drug that inhibits the growth of the microorganisms.

MBC (Minimal bactericidal concentration)
is the lowest concentration of a drug required to kill microorganisms.

Question 10

Define toxin and toxoid. Write down the difference between exotoxin and endotoxin. Mention 5 exotoxin and endotoxin producing bacteria.

Toxin
Toxins are the molecules that are produced by the microorganisms that may affect the cells in the infected host.

Toxoid
An inactivated toxin that has lost its ability to cause disease but has retained its immunogenicity. Releases from exotoxins.

Difference between exotoxin and endotoxin

Property	Exotoxin	Endotoxin
1. Source	Certain species of gram-positive and gram-negative bacteria	Cell wall of gram-negative bacteria
2. Secreted from cell	Yes	No
3. Chemistry	Polypeptide	Lipopolysaccharide
4. Location of genes	Plasmid or bacteriophage	Bacterial Chromosome
5. Toxicity	High	Low
6. Clinical effects	Various effects	Fever, Shock
10. Heat Stability	Destroyed rapidly at 60°C	Stable at 100°C for 1 hour
9. Vaccines	Toxoids are used as vaccines	No vaccine
11. Typical diseases	Tetanus, Botulism, Diphtheria	Meningococcemia, Sepsis by gram-negative rods

5 Exotoxin producing bacteria

1. *Staphylococcus aureus*
2. *Streptococcus pyogenes*
3. *Escherichia coli* (E. coli)
4. *Bacillus anthracis*
5. *Clostridium tetani*

5 Endotoxin producing bacteria

1. *Escherichia coli* (E. coli)
2. *Klebsiella*
3. *Salmonella*
4. *Shigella*
5. *Neisseria meningitidis*

Question 12

What is Selective toxicity and drug resistance? Name 4 beta-lactam drugs and 4 drugs that act by inhibiting protein synthesis. How do bacteria develop resistance to penicillin? Describe the method of development of drug resistance.

Selective toxicity
Selective inhibition of the growth of the microorganism without damage to the host. Achieved by exploiting the difference between the metabolism and structure of the microorganism and host cell.

Drug resistance
Drug resistance is the ability of a microbe to grow in the presence of a drug that would normally kill them or limit their growth.

4 beta-lactam drugs:

1. Penicillins
2. Cephalosporins
3. Carbapenems
4. Monobactams

4 drugs of protein synthesis inhibitors:

1. Clindamycin
2. Erythromycin
3. Azithromycin
4. Tetracycline

Development of resistance against penicillin:

1. Cleavage of β-lactam ring by β-lactamase.
2. Mutation in Penicillin-binding proteins (PBPs).
3. Mutation in porin proteins.

Method of Development of Drug Resistance

Mechanism	Important Example	Drugs Commonly Affected
Inactivate drug	Cleavage by β-lactamase	β-lactam drugs – Penicillin, Cephalosporins
Modify drug target in bacteria	1. Mutation in penicillin-binding protein	Penicillins
	2. Mutation in protein in 30S ribosomal subunit	Aminoglycosides, Streptomycin
	3. Replace alanine with lactate in peptidoglycan	Vancomycin
	4. Mutation in DNA gyrase	Quinolones
	5. Mutation in RNA polymerase	Rifampin
	6. Mutation in catalase-peroxidase	Isoniazid
Reduce permeability of drug	Mutation in porin proteins	Penicillins, Aminoglycosides, and others
Export of drug from bacteria	Multi-drug resistance pump	Tetracyclines, Sulfonamides, Quinolones

---

Question 13

What is the growth and growth curve of bacteria? Draw and label a bacterial growth curve. Mention in which phase exotoxin, endotoxin, and spores are produced.

Growth
It is an orderly increase in all the components of a bacteria.

Growth curve
It is the graphical representation of various stages of bacterial growth and multiplication in relation to time.

(Fig: Bacterial growth curve)
(Diagram showing X-axis as Time, Y-axis as Number of Bacteria. The curve shows four phases: Lag Phase, Log Phase, Stationary Phase, Death Phase.)

Phases:

1. **Lag phase:**
    - At this phase, bacteria prepare for multiplication.
    - No increase in the number of bacterial cells.
    - Antibiotics do not work well in this phase.
2. **Log phase:**
    - Bacteria remain in an actively multiplying state and divide very rapidly by binary fission.
    - Overwhelming infection occurs.
    - Antibiotics act well, because drugs are effective when cells are multiplying.
3. **Stationary phase:**
    - Bacterial multiplication and death becomes equal. There is no net increase or decrease in cell number.
    - Spore formation.
    - Exotoxin and endotoxin production.
4. **Death phase:**
    - Death of bacteria due to accumulation of a large amount of toxic product, and total deprivation of nutrition.
    - Exotoxins production by *C. diphtheriae*.

Exotoxin, endotoxins and spore production:
Exotoxin, Endotoxin, and Spores are produced in the **stationary phase**.

Short Notes on Key Topics

1. Trypticase Soya Broth

Trypticase soya broth is used in microbiology laboratories as a culture broth to grow aerobic bacteria. It is a complex general-purpose medium that is routinely used to grow certain pathogenic bacteria, which tend to have high nutritional requirements.

Preparation:
To prepare 1 liter of trypticase soya broth (TSB), the following ingredients are dissolved under gentle heat:

1. Trypticase peptone – 17 gm
2. Phytone peptone – 3 gm
3. Dipotassium phosphate – 2.5 gm
4. Dextrose – 2.5 gm
5. Distilled water – 1 liter

Adjustment of pH should be made using 1N HCl or 1N NaOH to reach a final target of pH 7.3 at 25°C. The solution is then autoclaved for 15 min at 121°C.

---

2. Bacterial Motility Test

Bacterial motility tests can be done in two methods:
A) MIU media
B) Hanging drop prep.

A) MIU media: In MIU media, if an organism is motile, then turbidity can be seen throughout the media.

- e.g., *E. Coli*, *Salmonella*.

If an organism is non-motile, then growth will be found along the stab line.

- e.g., *Shigella*, *Klebsiella*.

B) Hanging drop preparation:
i. Take some normal saline on a sterile slide.
ii. Then an isolated colony is dropped in the slide.
iii. Then petroleum jelly is added on the edges of a coverslip on one side.
iv. The coverslip is then placed on the slide, and the slide is mounted.
v. Mount the slide on the microscope at 10x and 40x respectively.
vi. The motile bacteria will move around while the non-motile will remain still.

---

3. Blood Culture

A blood culture is a test used to detect bacteria or fungi in a person's blood. Under normal conditions, the blood doesn't contain microorganisms. But their presence can indicate a bloodstream infection such as bacteremia or fungemia, which in severe cases may result in sepsis.

Indications:

1. Enteric fever
2. Septicemia
3. Pneumonia
4. Meningitis
5. Osteomyelitis
6. Brucellosis
7. Infective endocarditis
8. UTI
9. Pyrexia of unknown origins (PUO)

Media used for blood culture:

1. **Media for primary enrichment:**
   i. Tryptic soya broth.
2. **Media for sub-culture:**
   i. Blood agar media
   ii. MacConkey agar media
   iii. Chocolate agar media
   iv. Sabouraud Dextrose Agar (SDA) for fungal culture.

Different methods for blood culture:

1. **Conventional method or Traditional method:** Using liquid media.
2. **Lytic method:** In solid media.
3. **Automated method:** Very rapid (Within 6 hours).

Sample collection:

1. Clean the prick area by alcohol or antiseptic solution.
2. Wipe the same area repeatedly using an iodine solution.
3. The top of the culture bottle is disinfected using alcohol.
4. Draw about 10ml of blood with a disposable syringe (for adults). For neonates 1-2 ml and for infants and children 2-5 ml.
5. Change the needle of the syringe and inject the blood (about 10ml) into the blood culture bottle.

---

4. Bacterial Spore

Highly resistant structure formed under adverse environmental conditions such as nutritional depletion. Highly resistant to heat, dehydration, and chemicals etc.

Classification:

1. **Exospore:** Formed extracellularly from ends of the parent cell. (e.g., Fungus).
2. **Endospore:** Forms within the parent cell. (e.g., Bacteria).

Spore forming bacteria:

1. **Aerobic:** *Bacillus*.
2. **Anaerobic:** *Clostridium*.

Function of bacterial spore:

1. Spore form protects the bacteria from adverse environmental conditions.
2. Germinates in vegetative form when favourable conditions return.

Structure of spore:

1. Spore core
2. Spore wall
3. Cortex
4. Spore coat
5. Exosporium

(Diagram: Bacterial spore showing layers from inside out: Spore core, Spore wall, Cortex, Spore coat, Exosporium)

---

5. Plasmid

Extrachromosomal, double-stranded, circular DNA molecule that is capable of replication independently of the bacterial chromosome.

Type:

1. Transmissible plasmid
2. Non-transmissible plasmid

Function:

1. Employed in genetic engineering.
2. Plasmid carry the genes for:
   i. **Antibiotic resistance:** Mediated by a variety of enzymes.
   ii. **Exotoxins:** Exfoliative toxins of *Staphylococci* and Enterotoxin of *E. coli*.
   iii. **Pili:** Mediates the adherence of bacteria to epithelial cells.
   iv. **Resistance to heavy metals and UV light.**
   v. **Bacteriocin:** Toxic proteins produced by certain bacteria that are lethal for other bacteria.

---

6. Pasteurization

Heating of milk at such a temperature for such a period of time so that all viable pathogenic microorganisms are killed without significant alteration in flavor, taste or quality of milk.

Use of pasteurization:
Used to kill milk-borne pathogens:
i. *Mycobacterium bovis*
ii. *Streptococcus*
iii. *Brucella abortus*
iv. *Listeria*
v. *S. typhi*

Methods:

1. **Holding method:** 63°C for 30 min
2. **Flash method:** 72°C for 15 sec.
3. **Ultra high temperature (UHT):** 138°C for a fraction of a sec.

---

7. Toxoid

An inactive toxin that has lost its ability to cause disease but has retained its immunogenicity. Releases from exotoxins.

Properties:
An exotoxin has two main properties:

1. Toxicity
2. Antigenicity
   In a toxoid, toxicity is destroyed but antigenicity is preserved.

Production:
Toxoids are usually produced by heating the exotoxin. e.g., formalin.

Use:
Toxoids are used as a vaccine.

Example:

DPT

1. Diphtheria toxoid
2. Pertussis toxoid
3. Tetanus toxoid

---

8. Growth Curve

See answer to the question no. "13". Point starts - "Growth curve", point ends - "Death phase".

---

9. Drug Resistance

See answer to the question no. "12". Points - "Drug resistance", "Method of development of drug resistance".

Systemic Microbiology

Question 1: Classify Staphylococci on the basis of coagulase production. Name the lesions produced by staphylococcus. How will you diagnose the carbuncle of a patient in a laboratory?

Question 2: Short note on coagulase test, S. epidermidis, S. saprophyticus.

Question 3: Classify Streptococci. Mention the characteristics, toxins and enzymes of S. pyogenes. Mention the disease caused by S. pyogenes and lab diagnosis of it. Give the difference between streptococci and staphylococci.

Question 4: Short note on Aso titre, Toxic shock syndrome, Catalase test, Oxidase test, Biochemical tests.

Question 5: What is enterobacteriaceae? Give the common characteristics of it and classification of enterobacteriaceae. Name the lesions produce by them. Enumerate the causes and lab diagnosis of UTI.

Question 6: Name the causative organism of meningitis. What are the samples you will collect for meningitis? Lab diagnosis of bacterial meningitis. How will you differentiate bacterial, viral and tubercular meningitis in the lab?

Question 7: Mention the causes of sexual transmitted disease. Classify STIs. How will you diagnose a case of syphilis/gonorrhea/gonococcal urethritis in the lab?

Question 8: Mention the etiological agent, mode of transmission, pathogenesis and lab diagnosis of typhoid/enteric fever. Mention the interpretation of widal test.

Question 9: What is hospital acquired infection? Mention the common bacteria and disease produced by them in HAI. How can HAI be prevented?

Question 10: Short note on typhoid carrier and clinical features.

Question 11: Name important spore-forming anaerobic rod. Management of a patient with severe road traffic accident. How will you manage the patient to prevent tetanus? How will you diagnose the case of food poisoning caused by Clostridium botulinum?

Question 12: Mention the bacteria/organism causing diarrhoea. Mention the lab diagnosis of cholera and bacillary dysentery.

Question 13: Classify mycobacteria. Difference between primary and reinfection type of tuberculosis. Mention the lab diagnosis of pulmonary tuberculosis.

Question 1: Staphylococci

Classify Staphylococci on the basis of coagulase production. Name the lesions produced by staphylococcus. How will you diagnose the carbuncle of a patient in a laboratory?

Cocci are two types:

A. Gram-positive cocci: According to arrangement

1. **Grape-like cocci:** *Staphylococcus*
2. **In chain:** *Streptococcus*
3. **In pair:** *S. pneumoniae*

B. Gram-negative cocci:

1. **In pair:** *Neisseria*
2. **Other:** *Veillonella*

Characteristics of *Staphylococcus aureus*

A) Morphology and Staining characteristics:

1. Gram-positive, rounded in shape.
2. Arranged in grape-like clusters.
3. Non-motile.
4. Non-sporing.

B) Cultural characteristics:

1. Facultative anaerobe.
2. Optimum pH – 7.4.
3. Optimum temperature – 37°C.
4. Produces golden yellow colonies.

C) Growth characteristics:

1. **Blood agar:** Usually large colonies (pinhead) with a clear zone of beta-hemolysis.
2. **Nutrient agar:** Circular colony with golden yellow pigment.
3. **Nutrient broth:** Grows with uniform turbidity.
4. **Selective media (Mannitol Salt Agar):** Media with 10% NaCl inhibits most other flora but not *S. aureus*.

D) Antigenic Structure:

1. Protein-A
2. Cell wall peptidoglycan
3. Teichoic acid
4. Polysaccharide capsule
5. Cytoplasmic membrane

E) Enzymes:

1. **Catalase:** It converts H_2O_2 into H_2O and O_2.
2. **Coagulase:** Clumping factor.
3. **Phosphatase:** Produced by pathogenic strains.
4. **Hyaluronidase:** Spreading factor.
5. **Lipases:** Cleaves lipids.
6. **β-lactamase:** Splits the β-lactam ring of antibiotics (Penicillin).
7. **Deoxyribonuclease (DNase):** Hydrolysis of DNA.

F) Toxins and disease production (Superantigen):

Name of Toxin	Disease
1. Enterotoxin	1. Food poisoning
2. Toxic Shock Syndrome Toxin (TSST)	2. Toxic Shock Syndrome
3. Exfoliatin toxin	3. Scalded skin syndrome

G) Pathogenic criteria of *S. aureus*

1. Coagulase production.
2. Mannitol fermentation.
3. Hemolysis production.
4. Golden yellow pigment production.
5. Phosphatase production.

Classification of staphylococci on the basis of coagulase production:

- **Coagulase Positive:**
  - *Staphylococcus aureus*

- **Coagulase Negative:**
    - *Staphylococcus epidermidis*
    - *Staphylococcus saprophyticus*
    - Other less common species

Staphylococcal infection:

A. Due to invasion (Inflammation):

1. **Skin lesions:** Boils, Furuncle, Cellulitis, wound infection, Abscess, Paronychia, **Carbuncle**, Erysipelas.
2. **Lung infection:** Pneumonia, Lung abscess.
3. **CNS:** Meningitis, Brain abscess.
4. **Bone and joint:** Osteomyelitis, Arthritis.
5. **Heart:** Endocarditis.
6. **Septicemia.**

B. Due to toxins:

1. Food poisoning.
2. Toxic shock syndrome.
3. Scalded skin syndrome.

Lab diagnosis of staphylococcus aureus (from a carbuncle):

1. **Specimen:** Swab from lesion, Pus, exudate, blood, sputum, spinal fluid.
2. **Gram staining:** Gram-positive cocci, arranged in a cluster.
3. **Culture:**
    - **Blood agar:** Whitish colony / golden yellow colony with a clear zone of β-hemolysis.
    - **Nutrient agar:** Circular colony with golden yellow pigment.
4. **Biochemical tests:** Catalase test – Positive, Coagulase test – Positive.
5. **Serology test.**
6. **Phage typing.**

Question 2: Coagulase test & *S. epidermidis / S. saprophyticus*

*Short note on coagulase test, *S. epidermidis*, *S. saprophyticus*.

Coagulase test

Coagulase causes plasma to clot by activating the reactive coagulase factor in plasma to produce a thrombin-like substance which converts fibrinogen to fibrin.

- **Two methods:**
    - **Slide coagulase method:**
        1. A drop of normal saline is placed on a glass slide.
        2. A loopful of the colony of staphylococci is emulsified to form a smooth suspension.
        3. A loop of human or rabbit plasma is added and mixed.
        4. Coagulation of plasma within 15 seconds indicates that the staphylococcus is coagulase-positive.
           *This method detects **bound coagulase**.*

    - **Tube method:**
        1. Citrated human or rabbit plasma is diluted 1 in 10 with NS.
        2. 0.5 ml diluted plasma is placed in each of 2 small test tubes.
        3. Equal volume of overnight nutrient broth culture or growth from colonies on agar of the staphylococcus is mixed in one tube.
        4. In the second tube, sterile broth is included as a control.
        5. The tubes are incubated at 37°C. If a clot forms in 1-4 hours, the test is positive.
           *This method detects **free coagulase**.*

S. epidermidis

Staphylococcus epidermidis is coagulase-negative. Most strains form white colonies, but a few strains form pigmented colonies. It is a normal flora of the skin, respiratory, and gastrointestinal tracts. It can cause infection of prosthetic devices, and occasionally bacteremia and septicemia, particularly in immunosuppressed individuals.

S. saprophyticus

1. Skin commensal.
2. Coagulase-negative and is not usually differentiated from *S. epidermidis* clinically.
3. Causes urinary tract infection.
4. The organism is second to *E. coli* as a cause of community-acquired UTI in young women.

Question 3: Streptococci

Classify Streptococci. Mention the characteristics, toxins and enzymes of S. pyogenes. Mention the disease caused by S. pyogenes and lab diagnosis of it. Give the difference between streptococci and staphylococci.

Classification of Streptococci:

A. On the basis of hemolysis on blood agar:

- **β-hemolytic:** A clear zone around their colonies because of complete lysis of red blood cells. (e.g., *S. agalactiae*, *S. pyogenes*)
- **α-hemolytic:** Produce a greenish zone around their colonies as a result of incomplete lysis of red blood cells. (e.g., *S. pneumoniae*, Viridans streptococci)
- **γ- or non-hemolytic:** Colonies produce no hemolysis. (e.g., *S. bovis*, *Enterococcus faecalis*)

B. Based on C carbohydrate (Lancefield group):

Streptococci are grouped into A to H and K to U.

- **Group A:** *Streptococcus pyogenes*
- **Group B:** *Streptococcus agalactiae*

Important properties of *S. pyogenes*

1. Gram-positive cocci.
2. Arranged in chain form.
3. Catalase negative.
4. Non-motile.

Culture characteristics of *S. pyogenes*

1. Aerobic and facultative anaerobes.
2. Optimum pH: 7.4-7.6.
3. Optimum temperature: 37°C.
4. **On blood agar:** Circular, β-hemolytic colonies. Size of the colony: 1mm (Pinpoint).

Enzymes of *S. pyogenes*

1. **Streptokinase (Fibrinolysin):** Digests fibrin.
2. **Streptodornase:** Breaks down DNA, so decreases the viscosity of pus.
3. **Hyaluronidase:** It splits hyaluronic acid, an important component of the ground substance of connective tissue. That's why, when infection occurs it causes "spreading type" infection.

Toxins of *S. pyogenes*

1. **Erythrogenic toxin:** Associated with scarlet fever.
2. **Streptolysin O:** Oxygen labile.
3. **Streptolysin S:** Oxygen stable.
4. **Pyrogenic exotoxin B:** Is a protease which rapidly destroys tissue, causing necrotizing fasciitis.

Disease caused by *S. pyogenes*

1. **Due to invasion:** Cellulitis, necrotizing fasciitis, puerperal sepsis, bacterial endocarditis, meningitis.
2. **Due to local infection:** Pharyngitis, Sinusitis, Otitis, pneumonia.
3. **Toxigenic:** Scarlet fever, Toxic Shock Syndrome.
4. **Immune-mediated:**
    - **Acute glomerulonephritis:** Lab diagnosis - Anti-DNase B. (3-4 weeks after skin infection)
    - **Acute rheumatic fever:** Lab diagnosis - ASO titre. (3-4 weeks after streptococcal sore throat infection)

Lab diagnosis of *S. pyogenes*

1. **Specimen:** Swab from throat, Pus, Sputum, Blood.
2. **Microscopic examination:** Gram-positive cocci arranged in a long or short chain.
3. **Culture:** Shows pin-point colonies with a zone of complete hemolysis on blood agar media.
4. **Serology:** Estimation of ASO titre, Estimation of anti-DNase.

Difference between Staphylococcus and Streptococcus

Staphylococcus	Streptococcus
1. Infection is usually localized.	1. Infection is usually spreading.
2. Pus is thick.	2. Pus is thin due to the presence of DNase.
3. Morphology: Gram-positive cocci, arranged in irregular grape-like clusters.	3. Morphology: Gram-positive cocci, arranged often in a chain, pair, or single.
4. Culture: Grows in ordinary agar media.	4. Culture: Requires enriched media.
5. No growth on CVBA.	5. Grows on CVBA.
6. Grows in the presence of 10% NaCl.	6. Does not grow in the presence of NaCl.
7. Produces catalase.	7. Does not produce catalase.

Question 4: Biochemical Tests and Syndromes

Short note on ASO titre, Toxic shock syndrome, Catalase test, Oxidase test, Biochemical tests.

Catalase Test

- **Principle:** This test is used to differentiate those bacteria that produce the enzyme catalase, such as staphylococci, from non-catalase producing bacteria such as streptococci. Catalase acts as a catalyst in the breakdown of H_2O_2 to O_2 and H_2O. Bubbles of O_2 are released if the organism is a catalase producer.
- **Requirement:** 3% H_2O_2.
- **Method:**
    1. Take 2-3 ml of H_2O_2 into a test tube.

2. Take several colonies of the test organism using a sterile wooden stick or glass rod and immerse into the H_2O_2 solution.
3. Look for immediate bubble production.

- **Result:** Active bubbling → Positive catalase test. No bubbling → Negative catalase test.

Oxidase Test

- **Principle:** The oxidase test is to detect *Pseudomonas, Neisseria, Vibrio, Brucella* and *pasteurella* because all of which produce the enzyme cytochrome oxidase. A piece of filter paper is soaked with a few drops of oxidase reagent. When the organism is oxidase-producing, the phenylenediamine in the reagent will be oxidized to a deep purple color.
- **Method:**
    1. Place a piece of filter paper on a clean petri dish and add 2-3 drops of freshly prepared oxidase reagent.
    2. Using a stick of glass rod or wooden stick, pick a colony of the test organism and smear it on the filter paper.
    3. Look for the development of a deep purple color within a few seconds.

- **Result:** If a blue-purple color appears within 10 sec → oxidase positive. No color → Negative.

Biochemical Tests (Triple Sugar Iron Agar)

- **Composition:** Ferrous sulphate, peptone, three sugars (glucose, lactose, and sucrose), and phenol red.
- **Use:** The medium in the tube has a solid, poorly oxygenated area on the bottom called the "butt" and an angled, well-oxygenated area on top called the "slant". The organism is inoculated into the butt and across the surface of the slant to check for fermentation and H_2S production.

Butt (Acidic-A / Alkaline-K)	Slant	Gas (+)	H_2S	Organism Name
Acidic (A)	Acidic (A)	+	-	*E. coli, Klebsiella*
Acidic (A)	Alkaline (K)	-	-	*Shigella*
Alkaline (K)	Alkaline (K)	-	+	*Salmonella, Proteus*

Motility Indole Urea (MIU) Agar

- **Interpretation:**
    - **Turbidity** throughout the medium indicates a motile organism.
    - **Reddening** of the area at the neck of the tube indicates indole positive.
    - **Pink color** throughout the medium indicates urea positive.

Citrate Test

- **Citrate positive:** *Klebsiella, Serratia*.
- **Citrate negative:** *E. coli, Salmonella*.

ASO Titre
ASO titre means antibody against streptolysin-O. It is mainly produced 3-4 weeks after a recent streptococcal sore throat infection. The normal level of ASO titre is 160-200 IU/ml. When the ASO titre is higher than normal, it suggests a recent infection with *S. pyogenes*. The titre of ASO antibody can be important in the diagnosis of Rheumatic fever.

Toxic Shock Syndrome / Superantigen
It is mainly mediated by Staphylococci toxic shock syndrome toxin (TSST-1). It is manifested with high fever, shock, and erythematous skin rash. It occurs in tampon-using menstruating women and who use tight nasal packs after surgery for a prolonged period of time. Manifested by hypotension and fever. The TSST enters the blood stream and stimulates the release of a large amount of IL-1, IL-2 and tumor necrosis factors, which causes shock and toxemia that may lead to death.

Question 5: Enterobacteriaceae

What is Enterobacteriaceae? Give the common characteristics of it and classification of enterobacteriaceae. Name the lesions produced by them. Enumerate the causes and lab diagnosis of UTI.

Enterobacteriaceae
The family Enterobacteriaceae is composed of a heterogeneous group of closely related Gram-negative rods that are found in the soil, water, decaying materials, and the large bowel of humans and animals.

Common Characteristics:

1. Gram-negative bacilli.
2. Facultative anaerobes.
3. Motile or non-motile.
4. All ferment glucose with the production of acid and gas. Some are lactose fermenters and some are lactose non-fermenters.
5. Catalase positive.
6. Oxidase negative.
7. Reduce nitrates to nitrites.
8. Non-spore forming.
9. Some are capsulated.
10. Few are true pathogens; the majority are opportunistic pathogens.

Classification:

- **A. On the basis of lactose fermentation (using MacConkey agar media):**
    1. **Lactose fermenters:**
        - **Rapid:** *E. coli, Klebsiella, Enterobacter*.
        - **Slow:** *Citrobacter, Serratia, Edwardsiella*.

    2. **Lactose non-fermenters:** *Salmonella, Shigella, Proteus*.

- **B. On the basis of motility (Hanging drop, soft agar, MIU):**
    1. **Motile:** *E. coli, Salmonella, Proteus, Enterobacter, Serratia*.
    2. **Non-motile:** *Klebsiella, Shigella*.

- **C. On the basis of capsule:**
    1. **Capsulated:** *Klebsiella, S. Typhi*.
    2. **Non-capsulated:** *E. coli, Proteus, Enterobacter, Shigella, Serratia*.

- **D. On the basis of disease production:**
    1. **Normal flora:** All except *Salmonella, Shigella*.
    2. **Primary pathogen:** *Salmonella, Shigella*.
    3. **Opportunistic pathogen:** *E. coli, Proteus, Enterobacter, Citrobacter*.

Lesions produced:

Organism	Lesions/Diseases
E. coli	UTI, Diarrhea, Sepsis, Neonatal meningitis, Wound infection.
Klebsiella	UTI, Sepsis, Wound infection, Meningitis.

Salmonella	S. typhi - Typhoid fever; S. Paratyphi - Paratyphoid fever, Gastroenteritis.
Shigella	Bacillary dysentery.
Proteus	Wound infection.

Causes of UTI:

1. *E. coli* (80-90% of cases)
2. *S. saprophyticus* (young, newly married couple – Honeymoon cystitis)
3. *Klebsiella*
4. *Proteus*
5. *Pseudomonas*
6. *Acinetobacter*

Lab diagnosis of E. coli (from UTI):

- **A. Lesions outside the intestinal tract:**
    1. **Specimens:** Urine, pus, exudate, spinal fluid, blood culture.
    2. **Smear and staining:** A smear is prepared and Gram-staining is done to see the morphology.
    3. **Culture:** The specimen is inoculated on both blood and MacConkey agar media, incubated aerobically at 37°C overnight. Suspected colonies are studied for morphology and biochemical reactions.
    4. **Biochemical test:**
        - **TSI:** Both butt and slant are acidic.
        - **MIU:** Motile, Indole-positive, Urea-Negative.
        - **Citrate:** Negative.

Pathogenic / Diarrheagenic strains of *E. coli*

1. **Enteropathogenic *E. coli* (EPEC):** Diarrhea in infants.
2. **Enterotoxigenic *E. coli* (ETEC):** Traveller's/Watery diarrhea.
3. **Enteroaggregative *E. coli* (EAEC):** Acute and chronic diarrhea.
4. **Enteroinvasive *E. coli* (EIEC):** Shigella-like disease in children & travellers.
5. **Enterohemorrhagic *E. coli* (EHEC):** Hemorrhagic colitis & hemolytic uremic syndrome (HUS).

Question 6: Meningitis

Name the causative organism of meningitis. What are the samples you will collect for meningitis? Lab diagnosis of bacterial meningitis. How will you differentiate bacterial, viral and tubercular meningitis in the lab?

Name the causative agents of meningitis:

A. Bacterial cause:

1. First 6 weeks of life: Enteric gram-negative (*E. coli*, *Proteus* species), Group B streptococcus (*S. agalactiae*), *Listeria monocytogenes*.

2. 6 weeks to 6 years: *H. influenzae*, *N. meningitidis*, *S. pneumoniae*, *M. tuberculosis*.

3. 6 years to adult: *N. meningitidis*, *S. pneumoniae*, *M. tuberculosis*.

B. Viral causes: Polio virus, Echo virus, Coxsackie virus, Enterovirus, Rubella virus, Mumps virus, HSV-2 virus, Influenzae virus.

C. Fungal cause: *Cryptococcus neoformans*, *Candida albicans*.

Morphology of *N. meningitidis*

- Gram-negative cocci in pairs (diplococci).
- Kidney-shaped.
- Encapsulated.
- Non-motile.
- Non-spore forming.
- Remain intracellular within the polymorphonuclear cells and also remain extracellular.

Lab diagnosis of Meningitis (Bacterial):

- **Specimen:** CSF, Blood, Nasopharyngeal swab.
- **CSF examination:**
  - **First portion:** CSF is centrifuged and used to detect:
    i) Capsular antigen detection by latex agglutination test.
    ii) Biochemical analysis: increase CSF pressure, increase protein and decrease sugar.
    iii) Gram-staining: Pus cell with gram-negative diplococci.
  - **Second portion:**
    - Culture on chocolate agar media and incubate for 24-48 hours at 37°C under 5-10% CO_2.
    - At 24 hours of incubation - colonies are small, round, convex, gray, non-hemolytic, translucent.
  - **Third portion:**
    - Enriched broth BHI, granular turbidity is produced, then subculture in CA media.

- **Biochemical test:**
  - Oxidase test positive.

- Catalase test positive.

- **Serology:** Serogrouping by latex agglutination. Antibody to capsular antigen by ELISA.
- **Molecular diagnosis:** PCR.

Differentiation between bacterial, viral and tubercular meningitis:

	Normal	**Bacterial**	**Tubercular**	**Viral**
1. **Pressure**	60-100	Normal/Increased	Normal/Increased	Normal
2. **Cell count/mm³**	Lympho 0-5	Neutrophils often >1000-5000	Lympho 50-5000	Lympho 10-2000

3. **Appearance**	Clear	Turbid or purulent	Cobweb formation on standing	Clear
4. **Protein**	15-45 mg/dl	Increased 50-200	Increased 50-300	40-80
5. **Glucose**	40-80 mg/dl	Very low	Low	Normal
6. **Chloride**	710-750 mg/dl	Normal	<600	Normal
7. **Other findings**	Culture-sterile	Organism on Gram stain & culture	Tubercle bacilli on ZN stain, culture & PCR	Stain and culture no organism

---

Question 7: Sexually Transmitted Infections (STIs)

Mention the causes of sexual transmitted disease. Classify STIs. How will you diagnose a case of syphilis/gonorrhea/gonococcal urethritis in the lab?

Causes of Sexual Transmitted Disease:

Name of disease	Causative agent
1. Gonorrhea	*N. gonorrhoeae*
2. Syphilis	*Treponema pallidum*
3. AIDS	HIV

4. Genital herpes	HSV-2
5. Cervical cancer	Human papilloma virus
6. Trichomoniasis	*Trichomonas vaginalis*
7. Chlamydia	*Chlamydia trachomatis*

Classification of STI:

- **A. Bacterial causes:** *Neisseria gonorrhoeae*, *Treponema pallidum*, *Chlamydia trachomatis*.
- **B. Viral causes:** Human Immunodeficiency Virus, Herpes simplex virus-2, Human papilloma virus.
- **C. Fungal causes:** *Candida albicans*.
- **D. Protozoal causes:** *Trichomonas vaginalis*.

Lab diagnosis of Gonococcal Urethritis (*N. gonorrhoeae*)

- **Specimen collection:**
    - Acute case in male: Urethral discharge.
    - Chronic case in male: Prostatic smear by prostatic massage.
    - In case of female: Endocervical swab.
    - In case of infants: Eye swab.

- **Microscopy:** In acute cases, using Gram staining: Gram-negative diplococci, intracellular within the pus cell.
- **Culture:** If culture is delayed, it needs transport using Amies transport media.
    - **Non-Selective media:** Chocolate agar.
    - **Selective media:** Thayer Martin media - enriched media with lysed blood and antibiotics (Amphotericin-B, Vancomycin, trimethoprim, and colistin). Modified New York City media.
    - **Colony characteristics:** Translucent, grayish, piliated opaque.

- **Biochemical test:**
    - Oxidase test: Positive.
    - Glucose fermentation: Positive.

- **Serology:** Useful in chronic cases - Complement fixation test, Passive agglutination, Immunofluorescence test.

Difference between meningococcus and gonococcus:

Traits	Meningococci	Gonococci
Capsule	Present	Absent
Shape	Lens/half-moon shape (diplococci with side flattened)	Kidney shape (diplococci with side concave)
Fermentation	Glucose, maltose	Only glucose
Plasmid	Rarely present	Present
Colony	Circular	Varies in size with irregular margin
Habitat	Nasopharynx	Genital tract
Existence	Both intra & extracellular	Intracellular form

Question 8: Typhoid Fever (Enteric Fever)

Mention the etiological agent, mode of transmission, pathogenesis and lab diagnosis of typhoid/enteric fever. Mention the interpretation of the Widal test.

Etiological Agent Classification (according to pathogenicity):

1. **Salmonella that causes enteric fever:**
    - *Salmonella Typhi* → Typhoid fever
    - *Salmonella Paratyphi A, B, C* → Paratyphoid fever
2.
3. **Salmonella that causes food poisoning:**

    - *Salmonella Typhimurium*
    - *Salmonella Enteritidis*
4.
5. **Salmonella that causes septicemia:**
    - *Salmonella Choleraesuis*
6.

Morphology:

1. *Salmonella* are Gram-negative bacilli.
2. They are motile with the presence of peritrichous flagella.
3. They do not form capsules, spores.
4. Some strains of salmonella may produce fimbriae.
5. *Salmonella* are catalase positive and oxidase negative.
6. They ferment glucose.
7. They do not ferment lactose.

Antigenic Structure:
Salmonella possesses three major antigens:

1. **H or flagellar antigen**
2. **O or somatic antigen**
3. **Surface antigen - Vi antigen**

Mode of transmission:

1. Feco-oral transmission
2. Animal feces
3. Infected animal
4. Unwashed hand
5. Cross-contamination

Pathogenesis of Enteric Fever:

- **Route of entry:** Feco-oral, highly contaminated food and water.
- **Incubation period:** 7 to 14 days.
- **Infective dose:** In humans 10^3 - 10^6.

1. **Incubation stage:**
    - The ingested organisms contaminated with food and water are mostly destroyed in the stomach.
    - A sufficient number of bacilli pass through the gastric acid barrier and reach the duodenum and multiply in the alkaline medium.
    - In the small intestine, they are phagocytosed by neutrophils and macrophages in the submucosal coat (Peyer's patches).
    - The virulent bacilli resist intracellular killing and multiply within these cells.

- These cells enter the mesenteric lymph node.
- After multiplication, the bacilli invade the bloodstream via the thoracic duct.
- This causes transient bacteremia. Thus the internal organs are infected during primary bacteremia (First 7-10 days).
- During primary bacteremia there is no sign and symptom.
- The bloodstream is rapidly cleared off by the mononuclear phagocytic system (MPS) in the liver, bone marrow, and spleen.

2. **Septicemia stage:**
   - During primary bacteremia, the bacilli are able to live and multiply in MPS cells.
   - About the 10th day, the parasitized cells undergo necrosis.
   - From these internal organs, salmonella again enters into the blood. It is known as **secondary bacteremia**.
   - During this stage, the organism liberates endotoxin and produces signs and symptoms.

3. **Stage of localization:**
   - From the bloodstream, some organisms localize in organs, e.g., gall bladder, liver, spleen, bone marrow, and Peyer's patches.
   - The gall bladder is infected via the liver, resulting in cholecystitis.
   - The Peyer's patches become hyperplastic which may lead to necrosis of the superficial layer and sloughing of the affected follicles as typhoid ulcers, which may lead to hemorrhage and perforation.

Lab diagnosis:

- **Routine blood examination:** WBC count may be normal or low.
- **Isolation and identification of *S. typhi* and *S. paratyphi*:**
  - **Specimen:** Blood, Stool, Urine.
  - **Blood Culture:** Blood is collected taking strict aseptic measures. Culture in Tryptic soya broth (Conventional method) or lysis centrifugation method or Automated blood culture system. Blood needs to be subcultured from TSB on chocolate agar, blood agar and MacConkey agar media.
  - **Stool Culture:** At least two agar media (MacConkey and XLD) and one liquid media is used to culture stool. Any non-lactose fermenter colony observed needs to be identified using a biochemical test.

- **Urine Culture:** The urine is collected in a sterile container. The deposit is cultured on blood agar and MacConkey agar. The suspected colonies are studied using a biochemical test.

- **Biochemical test:**

TSI agar Butt / Slant	Gas	H_2S	Motility	Indole	Urea	Citrate	Interpretation
Acidic / Alkaline	+	+	+	-	-	+	*S. typhi*
Acidic / Alkaline	+	-	+	-	-	+	*S. paratyphi*

- **Widal test:** The patient serum is tested with "O" and "H" suspensions of *Salmonella typhi* / *paratyphi*.
- **ICT:** Patient serum is tested using a *S. typhi* and *paratyphi* specific ICT device.

Result of Widal test for typhoid fever:

TO Titre	TH Titre	Interpretation
1:80	1:40	Normal
1:160 or more	1:80	Typhoid fever
1:160 or more	1:160 or more	Typhoid fever
1:80	1:160 or more	Anamnestic reaction

Timeline of Diagnosis:

- **1st week:** Blood culture
- **2nd week:** Antibody titre by Widal test
- **3rd week:** Stool culture
- **4th week:** Urine culture

Question 9: Hospital Acquired Infections (HAI)

What is hospital acquired infection? Mention the common bacteria and diseases produced by them in HAI. How can HAI be prevented?

Hospital Acquired Infection (HAI)
It is an infection that is acquired in a hospital or other healthcare facilities, which appears after a time of at least 48 hours of hospitalization.

Mode of transmission:

1. **By direct contact:** Clothing, hand, etc.
2. **By airborne:** Droplet, dust, etc.
3. **By oral route:** Food, drinks.
4. **Parenteral route:** Blood transfusion, IV drip.

Type and causes of HAI:

Disease	Causative agent
1. UTI	*E. coli, Pseudomonas, Klebsiella, Proteus*.
2. LRTI	*H. influenzae, S. aureus, S. pneumoniae*, RSV.
3. Wound & skin infection	*S. aureus, E. coli, Proteus, S. epidermidis*, Anaerobic bacteria.
4. Burn	*Pseudomonas, S. aureus*, Group A streptococcus, *Acinetobacter*.
5. GIT infection	*Salmonella, Shigella, E. coli*.
6. Septicemia	Gram-negative bacilli, Viridans streptococci.

Prevention of HAI:

1. Isolation of the infected patient.
2. Proper sterilization and disinfection of the instruments and hospital inanimate objects which deal with the patient care.
3. Proper sanitation and disinfection of the hospital environment.
4. Disinfection and proper disposal of the excreta and infected materials from the hospital setup.
5. Administration of the proper antibiotics after proper drug sensitivity test.
6. Transmission routes should also be controlled by regular hand washing, disinfection of the equipment, and changing the cloth.
7. Administration of proper vaccine among the susceptible host.

Question 10: Typhoid Carrier State

Short note on typhoid carrier and clinical features.

Carrier of organism:

- About 2-4% of patients become chronic carriers.
- The bacilli persist in the gall bladder and are excreted in feces, called a **fecal carrier**.
- Persist in the kidney and are secreted in the urine, called a **urinary carrier**.

Types:

1. **Convalescent carrier:** Patient who continues to excrete typhoid bacilli in feces for 3 weeks to 3 months after clinical cure.
2. **Temporary carrier:** Those who shed the bacilli for >3 months but <1 year.
3. **Chronic carrier:** Those who excrete the bacilli for >1 year.

Clinical features:

- **First week:** Fever, headache, Myalgia, Bradycardia, Constipation.
    - **End of first week:** Rose spots, usually on the skin of the abdomen or chest; Splenomegaly; Cough; Abdominal distension with tenderness.

- **End of second week:** Bronchitis, Delirium.
- **Third week:** If untreated, toxemia increases and the patient may pass into a coma and die.

Complications of typhoid fever:

- **Bowel:** Perforation, Hemorrhage.
- **Septicemia:** Cholecystitis, Arthritis, Meningitis, Osteomyelitis.
- **Toxic phenomenon:** Myocarditis, Nephritis.

Question 11: Spore-Forming Rods (*Bacillus* & *Clostridium*)

Name some important spore-forming anaerobic rod. Management of a patient with a severe road traffic accident. How will you manage the patient to prevent tetanus? How will you diagnose the case of food poisoning caused by Clostridium botulinum?

Members of *Bacillus* species:

- **Pathogenic:** *B. anthracis*, *B. cereus*.
- **Non-pathogenic:** *B. subtilis*, *B. stearothermophilus* (to see the efficacy of autoclave).

General characteristics of *B. anthracis*

1. Gram-positive long rod with square ends.
2. Central spore.
3. Usually arranged in chains.
4. Non-motile.
5. Contains polypeptide capsule.
6. Aerobic.
7. Grows on ordinary media.

General characteristics of *C. tetani*

1. Gram-positive rods.
2. Spore-bearing.
3. Spore is a terminal which gives a "drum stick" or tennis racket appearance.
4. Anaerobic.
5. Motile.
6. Produces exotoxin: tetanospasmin.

Management of road traffic accident patient (to prevent tetanus):

1. **Wound care:** Exploration, cleaning, and debridement of the wound.
2. **Administration of antitoxin:** Immediate administration of TIG (Tetanus Immunoglobulin), 3000-10000 units in intravenous route.
3. **Tetanus toxoid:** Intramuscularly.
4. **Antibiotic:** Benzyl penicillin, 600 IU IV 6 hourly or metronidazole if the patient is allergic to penicillin.
5. **Anticonvulsant therapy:** Quiet room, Muscle relaxants, Sedatives (Diazepam IV).

Lab diagnosis of *C. botulinum* food poisoning:

1. Clinical suspicion.
2. Food history.
3. **Gram staining:** Gram-positive rod.
4. Toxin in leftover food and in blood is detected by the **mouse protection test**.
5. Mice are inoculated with a sample of the clinical specimen and will die unless protected by antitoxins.
6. ELISA.
7. PCR - Nucleic acid detection.

Question 12: Diarrhea and Dysentery

Mention the bacteria/organism causing diarrhea. Mention the lab diagnosis of cholera and bacillary dysentery.

Name of bacteria which cause diarrhea:

- **A. Non-invasive diarrhea:**
    1. Enterotoxigenic *E. coli*
    2. *V. cholerae*

3. *Clostridium difficile*
4. *Aeromonas*
5. *Plesiomonas*
6. *Yersinia enterocolitica*

-
- **B. Invasive diarrhea:**
    1. *Salmonella* species
    2. *Shigella dysenteriae*
    3. *Campylobacter* species
    4. Enteroinvasive *E. coli*, Enterohemorrhagic *E. coli*
    5. *V. parahaemolyticus*
-

Lab diagnosis of shigella / bacillary dysentery:

- **Specimen:** Fresh stool, mucous flecks, rectal swab.
- **Culture:** Specimens must be cultured with minimum delay.
    - **MacConkey / EMB / SS agar:** Form non-motile, lactose fermenting pale colored colonies. *S. sonnei* produces pink colonies after prolonged incubation.
    - **XLD agar:** Red pink colonies without a black center.
    - **DCA (Deoxycholate citrate agar):** Selective plating media for isolation of shigella from faeces, pale to pink colony.
- **Biochemical test:**
    - **TSI agar:** Alkaline slant and acidic butt. H_2S Negative. Gas production does not occur (sometimes +ve).
    - **MIU agar:** Non-motile.
    - **Citrate:** Negative.
    - **Oxidase:** Negative.

- **Confirmation of shigella by antiserum.**

Lab diagnosis of *V. cholerae*

- **Specimen:** Fresh stool, mucous flecks, vomitus, rectal swab.
- **Transport media:** Alkaline peptone water, Cary Blair media.
- **Microscopic examination:**
    - **Gram stain:** Gram-negative, comma-shaped bacilli.
    - **Dark ground illumination/microscopy:** Shooting star type motility.
    - **Hanging drop preparation:** Darting movement / Fish in stream appearance.

- **Culture:**
    - **MacConkey agar media:** Pale colonies (slow lactose fermenter).
    - **Blood agar media:** β hemolysis.

    - **Selective media (TCBS - Thiosulphate Citrate Bile Sucrose):** Large yellow colony.

- **Biochemical identification:**
    - **Oxidase:** Positive.
    - **Glucose:** Fermentation.

- **Confirmation by specific antiserum.**

Question 13: Mycobacteria and Tuberculosis

Classify mycobacteria. Difference between primary and reinfection type of tuberculosis. Mention the lab diagnosis of pulmonary tuberculosis.

Properties:

1. Mycobacteria are rod-shaped, slightly curved or straight, aerobic bacteria that don't form spores.
2. They stain with difficulty but once stained they resist decolorization with acid alcohol and so are called acid-fast bacilli.
3. M. tuberculosis causes tuberculosis, M. leprae causes leprosy.

Classification of mycobacteria (On the basis of disease production in human):

A. Pathogenic bacteria:

- **Mycobacterium tuberculosis complex:** *M. tuberculosis, M. bovis, M. microti, M. africanum*.
- **Mycobacterium leprae**.

B. Atypical mycobacteria / Non-tubercular mycobacteria:

- **Group-1 / Photochromogen:** *M. kansasii, M. marinum*.
- **Group-2 / Scotochromogen:** *M. gordonae, M. scrofulaceum*.
- **Group-3 / Non-chromogen:** *M. avium complex, M. gastri*.
- **Group-4 / Rapid grower:** *M. fortuitum chelonei complex, M. phlei*.

C. Non-pathogenic / saprophytic mycobacteria: *M. phlei, M. butyricum*.

Difference between primary and secondary TB:

Traits	Primary tuberculosis	Secondary tuberculosis
1. **Age of onset**	Children	Young adult
2. **Type of host**	Previously unexposed	Previously exposed
3. **Acquisition**	Inhalation, ingestion, transplacental	Reinfection, Reactivation
4. **Sites in lung**	Entire lung, but most commonly in lower lobe	Anywhere in lung, but most commonly in upper lobe
5. **Local lesion**	Small	Large
6. **Lymphatic involvement**	Happens commonly	Rarely
7. **Hematogenous dissemination**	Commonly	Rare
8. **Cavity formation**	No	Yes
9. **Tuberculin test**	Initially negative	Positive since the beginning
10. **Infectivity to others**	Rarely	Commonly

Lab diagnosis of pulmonary tuberculosis:

1. **Sample:** Sputum (3 consecutive samples), Gastric washing.
2. **Processing of sputum:**
   - Liquefaction - with N-acetyl L-cysteine.
   - Decontamination - with NaOH.
   - Neutralization - with buffer.
   - Concentration - by centrifugation.
3. **Microscopic examination:**

- **Ziehl-Neelsen stain:** M. tuberculosis appears as bright red, slender, straight or slightly curved rods against a blue/green background.
- **Auramine-Rhodamine stain:** AFB appears as bright yellow-orange color against a black background.

4. Culture: In solid media:

- **Lowenstein-Jensen media:** Incubate at 37°C with 5-10% CO_2 for up to 8 weeks. Buff, rough, tough colony of M. tuberculosis is seen.
- **Agar based media:** Middlebrook 7H10, Middlebrook 7H11.

5. Immunodiagnosis:

- Tuberculin skin test.
- Interferon-gamma release assay.

6. DNA detection:

- PCR.

7. Serology: Both IgG and IgM are detected by ELISA from patient serum.